

Volume of a Region in Space

If F is the constant function whose value is 1, then the sums in Equation (1) reduce to

$$S_n = \sum F(x_k, y_k, z_k) \Delta V_k = \sum 1 \cdot \Delta V_k = \sum \Delta V_k.$$

As Δx_k , Δy_k , and Δz_k approach zero, the cells ΔV_k become smaller and more numerous and fill up more and more of D . We therefore define the volume of D to be the triple integral

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \Delta V_k = \iiint_D dV.$$

DEFINITION The **volume** of a closed, bounded region D in space is

$$V = \iiint_D dV.$$

EXAMPLE 1 Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Solution The volume is

$$V = \iiint_D dz \, dy \, dx,$$

the integral of $F(x, y, z) = 1$ over D . To find the limits of integration for evaluating the integral, we first sketch the region. The surfaces (Figure 15.30) intersect on the elliptical cylinder $x^2 + 3y^2 = 8 - x^2 - y^2$ or $x^2 + 2y^2 = 4$, $z > 0$. The boundary of the region R , the projection of D onto the xy -plane, is an ellipse with the same equation: $x^2 + 2y^2 = 4$. The “upper” boundary of R is the curve $y = \sqrt{(4 - x^2)/2}$. The lower boundary is the curve $y = -\sqrt{(4 - x^2)/2}$.

Now we find the z -limits of integration. The line M passing through a typical point (x, y) in R parallel to the z -axis enters D at $z = x^2 + 3y^2$ and leaves at $z = 8 - x^2 - y^2$.

Next we find the y -limits of integration. The line L through (x, y) parallel to the y -axis enters R at $y = -\sqrt{(4 - x^2)/2}$ and leaves at $y = \sqrt{(4 - x^2)/2}$.

Finally we find the x -limits of integration. As L sweeps across R , the value of x varies from $x = -2$ at $(-2, 0, 0)$ to $x = 2$ at $(2, 0, 0)$. The volume of D is

$$\begin{aligned} V &= \iiint_D dz \, dy \, dx \\ &= \int_{-2}^2 \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} \int_{x^2+3y^2}^{8-x^2-y^2} dz \, dy \, dx \\ &= \int_{-2}^2 \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} (8 - 2x^2 - 4y^2) \, dy \, dx \\ &= \int_{-2}^2 \left[(8 - 2x^2)y - \frac{4}{3}y^3 \right]_{y=-\sqrt{(4-x^2)/2}}^{y=\sqrt{(4-x^2)/2}} dx \\ &= \int_{-2}^2 \left(2(8 - 2x^2)\sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx \\ &= \int_{-2}^2 \left[8 \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right] dx = \frac{4\sqrt{2}}{3} \int_{-2}^2 (4 - x^2)^{3/2} dx \\ &= 8\pi\sqrt{2}. \end{aligned}$$

After integration with the substitution $x = 2 \sin u$

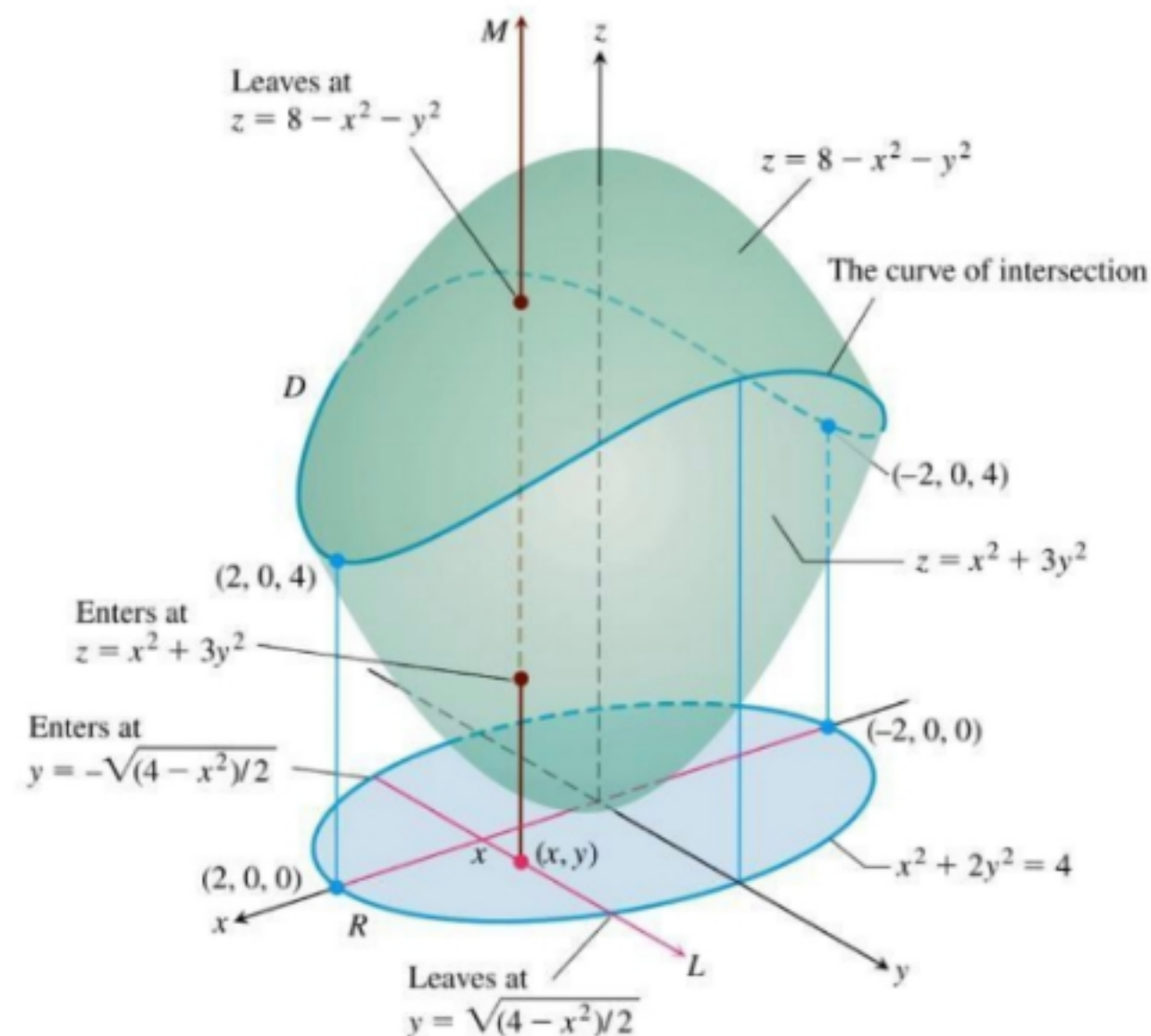


FIGURE 15.30 The volume of the region enclosed by two paraboloids, calculated in Example 1.

EXAMPLE 2 Set up the limits of integration for evaluating the triple integral of a function $F(x, y, z)$ over the tetrahedron D with vertices $(0, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$, and $(0, 1, 1)$. Use the order of integration $dy \, dz \, dx$.

Solution We sketch D along with its “shadow” R in the xz -plane (Figure 15.31). The upper (right-hand) bounding surface of D lies in the plane $y = 1$. The lower (left-hand) bounding surface lies in the plane $y = x + z$. The upper boundary of R is the line $z = 1 - x$. The lower boundary is the line $z = 0$.

First we find the y -limits of integration. The line through a typical point (x, z) in R parallel to the y -axis enters D at $y = x + z$ and leaves at $y = 1$.

Next we find the z -limits of integration. The line L through (x, z) parallel to the z -axis enters R at $z = 0$ and leaves at $z = 1 - x$.

Finally we find the x -limits of integration. As L sweeps across R , the value of x varies from $x = 0$ to $x = 1$. The integral is

$$\int_0^1 \int_0^{1-x} \int_{x+z}^1 F(x, y, z) \, dy \, dz \, dx.$$

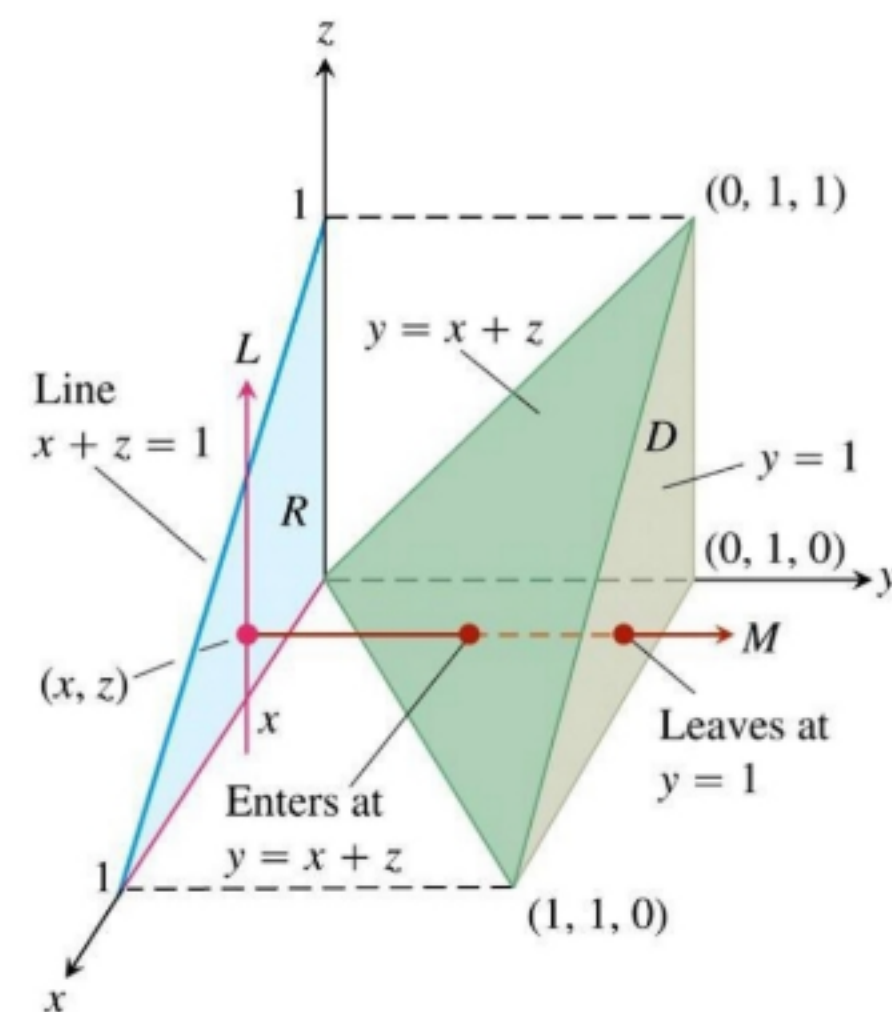


FIGURE 15.31 Finding the limits of integration for evaluating the triple integral of a function defined over the tetrahedron D (Examples 2 and 3).

Average Value of a Function in Space

The average value of a function F over a region D in space is defined by the formula

$$\text{Average value of } F \text{ over } D = \frac{1}{\text{volume of } D} \iiint_D F \, dV. \quad (2)$$

For example, if $F(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, then the average value of F over D is the average distance of points in D from the origin. If $F(x, y, z)$ is the temperature at (x, y, z) on a solid that occupies a region D in space, then the average value of F over D is the average temperature of the solid.

EXAMPLE 4 Find the average value of $F(x, y, z) = xyz$ throughout the cubical region D bounded by the coordinate planes and the planes $x = 2$, $y = 2$, and $z = 2$ in the first octant.

Solution We sketch the cube with enough detail to show the limits of integration (Figure 15.33). We then use Equation (2) to calculate the average value of F over the cube.

The volume of the region D is $(2)(2)(2) = 8$. The value of the integral of F over the cube is

$$\begin{aligned} \int_0^2 \int_0^2 \int_0^2 xyz \, dx \, dy \, dz &= \int_0^2 \int_0^2 \left[\frac{x^2}{2} yz \right]_{x=0}^{x=2} dy \, dz = \int_0^2 \int_0^2 2yz \, dy \, dz \\ &= \int_0^2 \left[y^2 z \right]_{y=0}^{y=2} dz = \int_0^2 4z \, dz = \left[2z^2 \right]_0^2 = 8. \end{aligned}$$

With these values, Equation (2) gives

$$\text{Average value of } xyz \text{ over the cube} = \frac{1}{\text{volume}} \iiint_{\text{cube}} xyz \, dV = \left(\frac{1}{8} \right) (8) = 1.$$

In evaluating the integral, we chose the order $dx \, dy \, dz$, but any of the other five possible orders would have done as well. ■

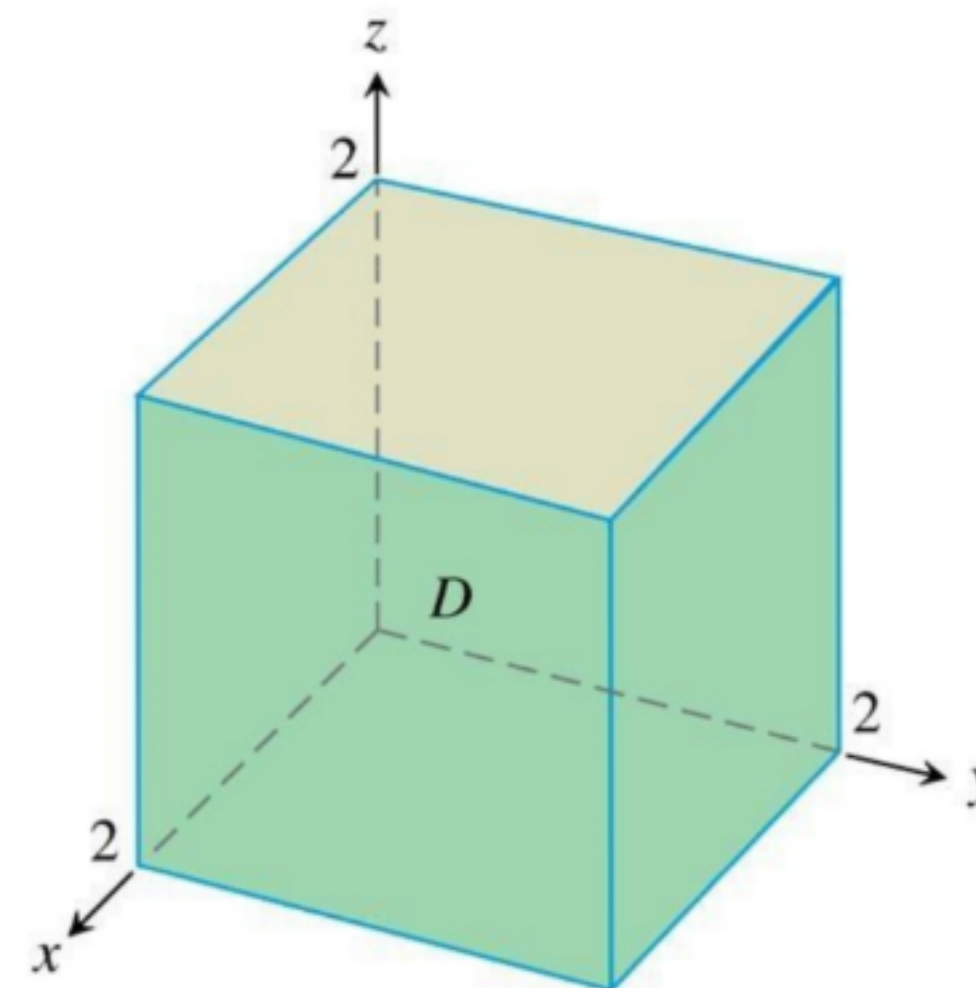


FIGURE 15.33 The region of integration in Example 4.

EXAMPLE 1 Find the center of mass of a solid of constant density δ bounded below by the disk $R: x^2 + y^2 \leq 4$ in the plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$ (Figure 15.35).

Solution By symmetry $\bar{x} = \bar{y} = 0$. To find $\bar{z}$, we first calculate

$$\begin{aligned} M_{xy} &= \iiint_R \int_{z=0}^{z=4-x^2-y^2} z \delta \, dz \, dy \, dx = \iint_R \left[\frac{z^2}{2} \right]_{z=0}^{z=4-x^2-y^2} \delta \, dy \, dx \\ &= \frac{\delta}{2} \iint_R (4 - x^2 - y^2)^2 \, dy \, dx \\ &= \frac{\delta}{2} \int_0^{2\pi} \int_0^2 (4 - r^2)^2 r \, dr \, d\theta \quad \text{Polar coordinates simplify the integration.} \\ &= \frac{\delta}{2} \int_0^{2\pi} \left[-\frac{1}{6} (4 - r^2)^3 \right]_{r=0}^{r=2} d\theta = \frac{16\delta}{3} \int_0^{2\pi} d\theta = \frac{32\pi\delta}{3}. \end{aligned}$$

A similar calculation gives the mass

$$M = \iiint_R \int_0^{4-x^2-y^2} \delta \, dz \, dy \, dx = 8\pi\delta.$$

Therefore $\bar{z} = (M_{xy}/M) = 4/3$ and the center of mass is $(\bar{x}, \bar{y}, \bar{z}) = (0, 0, 4/3)$.

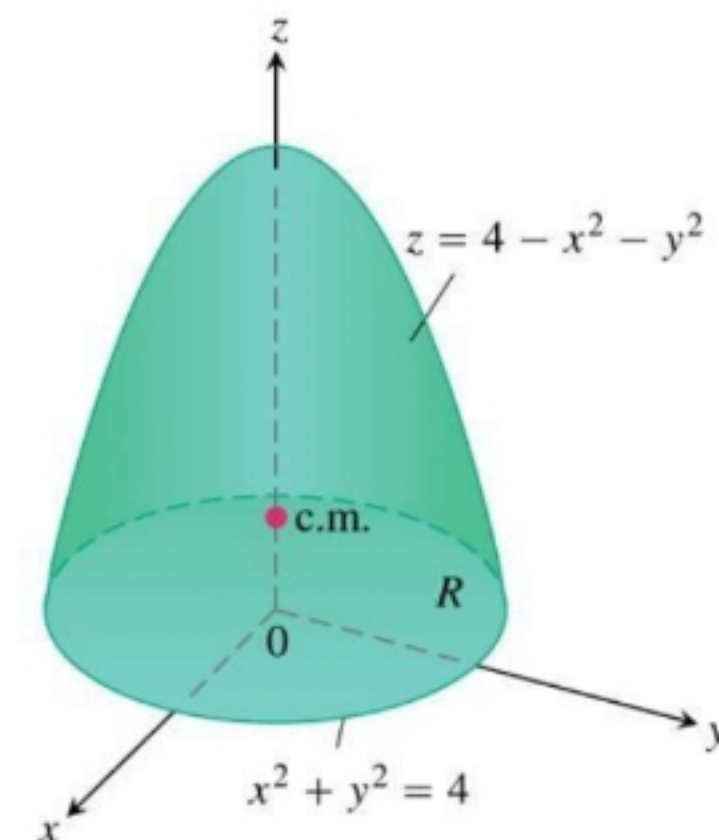


FIGURE 15.35 Finding the center of mass of a solid (Example 1).

TABLE 15.1 Mass and first moment formulas

THREE-DIMENSIONAL SOLID

Mass: $M = \iiint_D \delta \, dV$ $\delta = \delta(x, y, z)$ is the density at (x, y, z) .

First moments about the coordinate planes:

$$M_{yz} = \iiint_D x \, \delta \, dV, \quad M_{xz} = \iiint_D y \, \delta \, dV, \quad M_{xy} = \iiint_D z \, \delta \, dV$$

Center of mass:

$$\bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}$$

TWO-DIMENSIONAL PLATE

Mass: $M = \iint_R \delta \, dA$ $\delta = \delta(x, y)$ is the density at (x, y) .

First moments: $M_y = \iint_R x \, \delta \, dA, \quad M_x = \iint_R y \, \delta \, dA$

Center of mass: $\bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}$

TABLE 15.2 Moments of inertia (second moments) formulas

THREE-DIMENSIONAL SOLID

About the x-axis: $I_x = \iiint (y^2 + z^2) \, \delta \, dV$ $\delta = \delta(x, y, z)$

About the y-axis: $I_y = \iiint (x^2 + z^2) \, \delta \, dV$

About the z-axis: $I_z = \iiint (x^2 + y^2) \, \delta \, dV$

About a line L : $I_L = \iiint r^2 \, \delta \, dV$ $r(x, y, z)$ = distance from the point (x, y, z) to line L

TWO-DIMENSIONAL PLATE

About the x-axis: $I_x = \iint y^2 \, \delta \, dA$ $\delta = \delta(x, y)$

About the y-axis: $I_y = \iint x^2 \, \delta \, dA$

About a line L : $I_L = \iint r^2(x, y) \, \delta \, dA$ $r(x, y)$ = distance from (x, y) to L

About the origin (polar moment): $I_0 = \iint (x^2 + y^2) \, \delta \, dA = I_x + I_y$

EXAMPLE 2 Find the centroid of the region in the first quadrant that is bounded above by the line $y = x$ and below by the parabola $y = x^2$.

Solution We sketch the region and include enough detail to determine the limits of integration (Figure 15.36). We then set δ equal to 1 and evaluate the appropriate formulas from Table 15.1:

$$M = \int_0^1 \int_{x^2}^x 1 \, dy \, dx = \int_0^1 \left[y \right]_{y=x^2}^{y=x} dx = \int_0^1 (x - x^2) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{6}$$

$$\begin{aligned} M_x &= \int_0^1 \int_{x^2}^x y \, dy \, dx = \int_0^1 \left[\frac{y^2}{2} \right]_{y=x^2}^{y=x} dx \\ &= \int_0^1 \left(\frac{x^2}{2} - \frac{x^4}{2} \right) dx = \left[\frac{x^3}{6} - \frac{x^5}{10} \right]_0^1 = \frac{1}{15} \end{aligned}$$

$$M_y = \int_0^1 \int_{x^2}^x x \, dy \, dx = \int_0^1 \left[xy \right]_{y=x^2}^{y=x} dx = \int_0^1 (x^2 - x^3) \, dx = \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{12}.$$

From these values of M , M_x , and M_y , we find

$$\bar{x} = \frac{M_y}{M} = \frac{1/12}{1/6} = \frac{1}{2} \quad \text{and} \quad \bar{y} = \frac{M_x}{M} = \frac{1/15}{1/6} = \frac{2}{5}.$$

The centroid is the point $(1/2, 2/5)$. ■

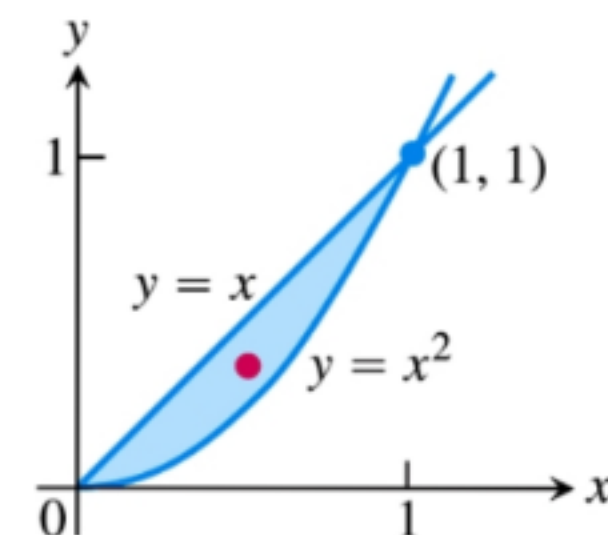


FIGURE 15.36 The centroid of this region is found in Example 2.

EXAMPLE 3 Find I_x, I_y, I_z for the rectangular solid of constant density δ shown in Figure 15.39.

Solution The formula for I_x gives

$$I_x = \int_{-c/2}^{c/2} \int_{-b/2}^{b/2} \int_{-a/2}^{a/2} (y^2 + z^2) \delta \, dx \, dy \, dz.$$

We can avoid some of the work of integration by observing that $(y^2 + z^2)\delta$ is an even function of x, y , and z since δ is constant. The rectangular solid consists of eight symmetric pieces, one in each octant. We can evaluate the integral on one of these pieces and then multiply by 8 to get the total value.

$$\begin{aligned} I_x &= 8 \int_0^{c/2} \int_0^{b/2} \int_0^{a/2} (y^2 + z^2) \delta \, dx \, dy \, dz = 4a\delta \int_0^{c/2} \int_0^{b/2} (y^2 + z^2) \, dy \, dz \\ &= 4a\delta \int_0^{c/2} \left[\frac{y^3}{3} + z^2 y \right]_{y=0}^{y=b/2} dz \\ &= 4a\delta \int_0^{c/2} \left(\frac{b^3}{24} + \frac{z^2 b}{2} \right) dz \\ &= 4a\delta \left(\frac{b^3 c}{48} + \frac{c^3 b}{48} \right) = \frac{abc\delta}{12} (b^2 + c^2) = \frac{M}{12} (b^2 + c^2). \quad M = abc\delta \end{aligned}$$

Similarly,

$$I_y = \frac{M}{12} (a^2 + c^2) \quad \text{and} \quad I_z = \frac{M}{12} (a^2 + b^2).$$

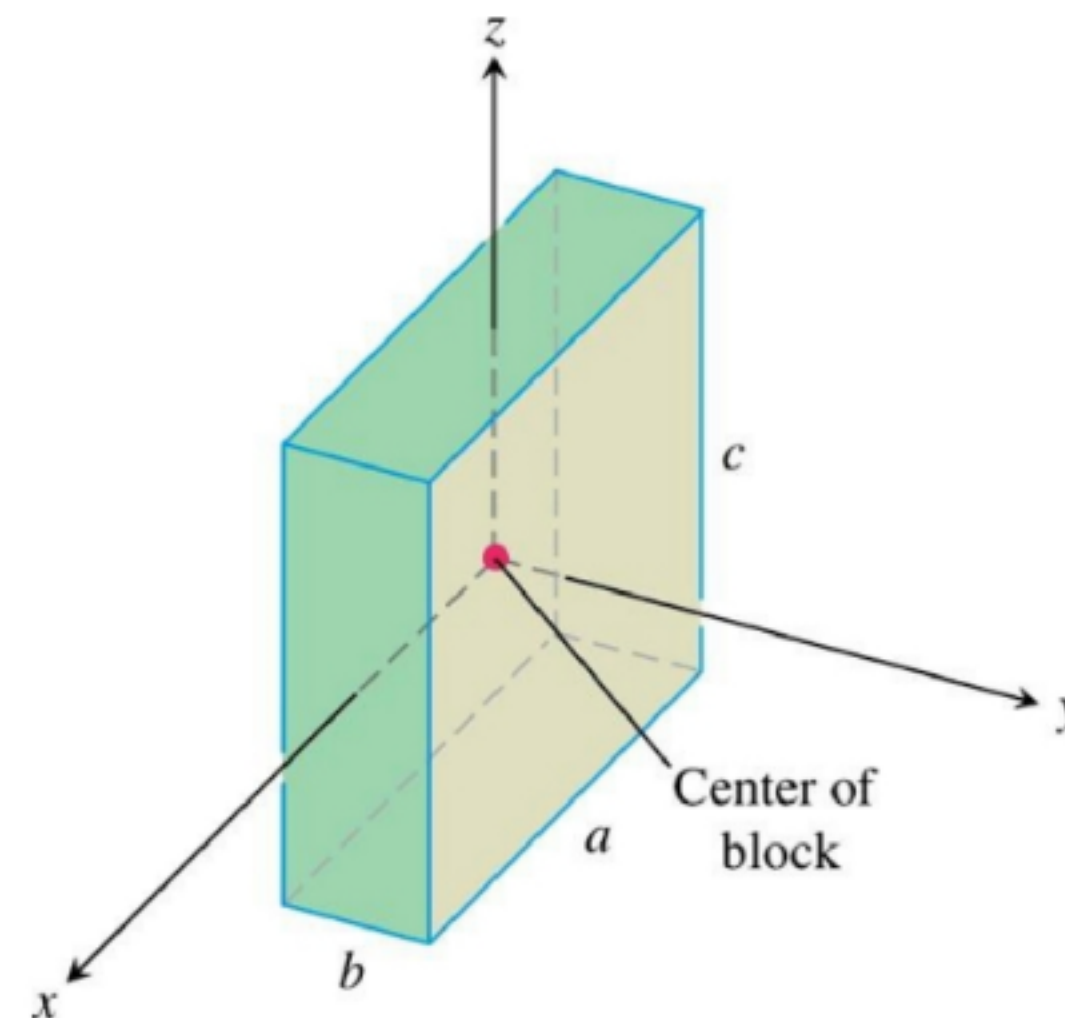


FIGURE 15.39 Finding I_x, I_y , and I_z for the block shown here. The origin lies at the center of the block (Example 3).

EXAMPLE 4 A thin plate covers the triangular region bounded by the x -axis and the lines $x = 1$ and $y = 2x$ in the first quadrant. The plate's density at the point (x, y) is $\delta(x, y) = 6x + 6y + 6$. Find the plate's moments of inertia about the coordinate axes and the origin.

Solution We sketch the plate and put in enough detail to determine the limits of integration for the integrals we have to evaluate (Figure 15.41). The moment of inertia about the x -axis is

$$\begin{aligned} I_x &= \int_0^1 \int_0^{2x} y^2 \delta(x, y) \, dy \, dx = \int_0^1 \int_0^{2x} (6xy^2 + 6y^3 + 6y^2) \, dy \, dx \\ &= \int_0^1 \left[2xy^3 + \frac{3}{2}y^4 + 2y^3 \right]_{y=0}^{y=2x} dx = \int_0^1 (40x^4 + 16x^3) \, dx \\ &= \left[8x^5 + 4x^4 \right]_0^1 = 12. \end{aligned}$$

Evaluating Triple Iterated Integrals

Evaluate the integrals in Exercises 7–20.

7. $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$
8. $\int_0^{\sqrt{2}} \int_0^{3y} \int_{x^2+3y^2}^{8-x^2-y^2} dz dx dy$ 9. $\int_1^e \int_1^{e^2} \int_1^{e^3} \frac{1}{xyz} dx dy dz$
10. $\int_0^1 \int_0^{3-3x} \int_0^{3-3x-y} dz dy dx$ 11. $\int_0^{\pi/6} \int_0^1 \int_{-2}^3 y \sin z dx dy dz$
12. $\int_{-1}^1 \int_0^1 \int_0^2 (x + y + z) dy dx dz$

Average Values

In Exercises 37–40, find the average value of $F(x, y, z)$ over the given region.

37. $F(x, y, z) = x^2 + 9$ over the cube in the first octant bounded by the coordinate planes and the planes $x = 2$, $y = 2$, and $z = 2$
38. $F(x, y, z) = x + y - z$ over the rectangular solid in the first octant bounded by the coordinate planes and the planes $x = 1$, $y = 1$, and $z = 2$
39. $F(x, y, z) = x^2 + y^2 + z^2$ over the cube in the first octant bounded by the coordinate planes and the planes $x = 1$, $y = 1$, and $z = 1$

Plates of Constant Density

1. **Finding a center of mass** Find the center of mass of a thin plate of density $\delta = 3$ bounded by the lines $x = 0$, $y = x$, and the parabola $y = 2 - x^2$ in the first quadrant.
2. **Finding moments of inertia** Find the moments of inertia about the coordinate axes of a thin rectangular plate of constant density δ bounded by the lines $x = 3$ and $y = 3$ in the first quadrant.
3. **Finding a centroid** Find the centroid of the region in the first quadrant bounded by the x -axis, the parabola $y^2 = 2x$, and the line $x + y = 4$.
4. **Finding a centroid** Find the centroid of the triangular region cut from the first quadrant by the line $x + y = 3$.

Plates with Varying Density

11. **Finding a moment of inertia** Find the moment of inertia about the x -axis of a thin plate bounded by the parabola $x = y - y^2$ and the line $x + y = 0$ if $\delta(x, y) = x + y$.
12. **Finding mass** Find the mass of a thin plate occupying the smaller region cut from the ellipse $x^2 + 4y^2 = 12$ by the parabola $x = 4y^2$ if $\delta(x, y) = 5x$.
13. **Finding a center of mass** Find the center of mass of a thin triangular plate bounded by the y -axis and the lines $y = x$ and $y = 2 - x$ if $\delta(x, y) = 6x + 3y + 3$.